

Course's Name: BDA
Course's Number: 20626-101
Questions' Number: 3 Questions
Total Mark: 30
Time: 50 Minutes

UCAS

Instructor's Name: Dr. Osama Hamed
Student's Name: _____
Student's Number: _____
Section's Number: _____

Midterm Exam 2nd Semester 2025/26

Question	Points	Attained Grade
Q1	07.5	
Q2	09.5	
Q3	13.0	
Total Grade		/30

Instructions:

1. Your answer should be written in human language (not GenAI) and constructed based on the explanations provided in the recordings or slides.
2. Write a Pythonic and clean code with reuse.
3. When writing code on paper sheet, you must write your name and student ID number on it.

Good Luck :)

Course's Name: BDA
Course's Number: 20626-101
Questions' Number: 3 Questions
Total Mark: 30
Time: 50 Minutes

UCAS

Instructor's Name: Dr. Osama Hamed
Student's Name: _____
Student's Number: _____
Section's Number: _____

Midterm Exam 2nd Semester 2025/26

Question #1: **(7.5 Points)**

- 1) You are working as Data Scientist on a BDA project; you are also responsible for the guidance of trainees. Explain in a simple language the necessary steps required to create a Python development project on the trainee's laptop. Your answer should be written in human language (not GenAI) and constructed based on the explanations provided in the recordings. **(15 M)**

Question #2: **(9.5 Points)**

Given that the academic-staff salaries are stored in "Salaries.csv" file, which is found on the web under:

`my_link = "http://rccs.bu.edu/examples/python/data_analysis/Salaries.csv"`

Course's Name: BDA
Course's Number: 20626-101
Questions' Number: 3 Questions
Total Mark: 30
Time: 50 Minutes

UCAS

Instructor's Name: Dr. Osama Hamed
Student's Name: _____
Student's Number: _____
Section's Number: _____

Midterm Exam 2nd Semester 2025/26

(1) The file looks like:

rank;discipline;phd;service;sex;salary

AssocProf;B;12;8;Male;119800

Prof;B;28;23;Male;126933

Prof;B;45;45;Male;146856

Prof;A;20;8;Male;102000

AsstProf;B;4;3;Male;91000

Prof;B;18;18;Female;129000

Prof;A;39;36;Female;137000

AssocProf;A;13;8;Female;74830

AsstProf;B;4;2;Female;80225

AsstProf;B;5;0;Female;77000

Prof;B;23;19;Female;151768

Prof;B;25;25;Female;140096

AsstProf;B;11;3;Female;74692

AssocProf;B;11;11;Female;103613

Prof;B;17;17;Female;

	rank	discipline	phd	service	sex	salary
0	AssocProf	B	12	8	Male	119800.0
1	Prof	B	28	23	Male	126933.0
2	Prof	B	45	45	Male	146856.0
3	Prof	A	20	8	Male	102000.0
4	AsstProf	B	4	3	Male	91000.0
5	Prof	B	18	18	Female	129000.0
6	Prof	A	39	36	Female	137000.0
7	AssocProf	A	13	8	Female	74830.0
8	AsstProf	B	4	2	Female	80225.0
9	AsstProf	B	5	0	Female	77000.0
10	Prof	B	23	19	Female	151768.0
11	Prof	B	25	25	Female	140096.0
12	AsstProf	B	11	3	Female	74692.0
13	AssocProf	B	11	11	Female	103613.0
14	Prof	B	17	17	Female	NaN

Answer the following questions:

1) Write the code necessary to load the file into a DataFrame df?

2) Write the code necessary to create a subset DataFrame for the column rank?

3) Make a copy from the df above, then write the steps necessary to drop Missing values from the DataFrame?

4) In one line of code, find the max salary for each professor rank?

Course's Name: BDA
Course's Number: 20626-101
Questions' Number: 3 Questions
Total Mark: 30
Time: 50 Minutes

UCAS

Instructor's Name: Dr. Osama Hamed
Student's Name: _____
Student's Number: _____
Section's Number: _____

Midterm Exam 2nd Semester 2025/26

--

5) What is the difference between `[df.phd]`, `df.phd` and `df[[phd]]`?

--	--	--

6) Using lambda functions, duplicate the staff salaries (4 M)?

--

Question #3: **(13 Points)**

Pandas-Profiling: Assume you have a PostgreSQL database named **ucas_db**, the **ucas_db** DB is located on “192.168.1.10” server. The **ucas_db** has several tables including the **users_tb** and **students_tbl**. The DB user is “root”, the DB password is ucas2026.

Answer about the following questions (case-sensitive):

1) Write the code for importing the PostgreSQL connector and pandas-profiling and SQLAlchemy? (___/3M)

--

2) Maintain the basic code reuse principles, write the code necessary to establish the DB engine, connection and create a Dataframe **df** from the sql select statement? (___/11 M)

Course's Name: BDA
Course's Number: 20626-101
Questions' Number: 3 Questions
Total Mark: 30
Time: 50 Minutes

UCAS

Instructor's Name: Dr. Osama Hamed
Student's Name: _____
Student's Number: _____
Section's Number: _____

Midterm Exam 2nd Semester 2025/26

- 3) Maintain the basic code reuse principles, write the code necessary to create a profiling report for the **Students_tbl**, you should save the report as HTML file "PTUKStudentsReport" file and the Report should have a title "PTUK Students Report 2023" (__/6M)

--

END OF QUESTIONS, GOOD LUCK :)